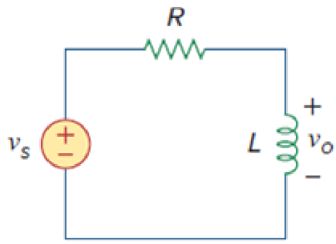
Chapter 14, Problem 1PP

Problem

Obtain the transfer function $\mathbf{V}_0/\mathbf{V}_s$ of the *RL* circuit in Fig. 14.4, assuming $v_s = V_m \cos \omega t$. Sketch its frequency response.

Figure 14.4: *RL* circuit



Step-by-step solution

Step 1 of 8

Refer to Figure 14.4 in the textbook.

Draw the frequency – domain equivalent circuit.

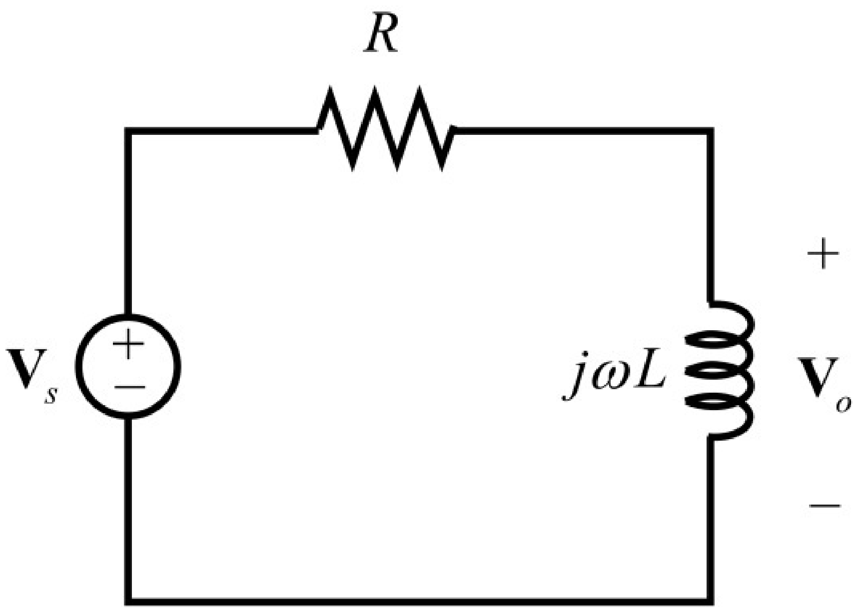


Figure 1

Step 2 of 8

Apply voltage division rule to write the output voltage expression.

$$\mathbf{V}_o = \left(\frac{j\omega L}{j\omega L + R}\right)\mathbf{V}_s$$

$$\frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{j\omega L}{j\omega L + R}$$

$$H(\omega) = \frac{j\omega L}{j\omega L + R}$$

Thus, the transfer function of the RL circuit is $\boxed{\frac{j\omega L}{j\omega L + R}}$.

Step 3 of 8

Rewrite the transfer function in standard form.

$$\begin{aligned} H(\omega) &= \frac{j\omega L}{j\omega L + R} \\ &= \frac{\frac{j\omega L}{R}}{1 + \frac{j\omega L}{R}} \\ &= \frac{j\left(\frac{\omega}{\omega_0}\right)}{1 + j\left(\frac{\omega}{\omega_0}\right)} \end{aligned}$$

Here $\omega_0 = \frac{R}{L}$.

Calculate the magnitude of the transfer function.

$$H = \frac{\left(\frac{\omega}{\omega_0}\right)}{\sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}}$$

Step 4 of 8

Calculate the magnitude at different points.

ω	H
0	0
ω_0	0.707
$2\omega_0$	0.894
$3\omega_0$	0.948
$4\omega_0$	0.97
$5\omega_0$	0.98
$10\omega_0$	0.995
$20\omega_0$	0.998
$50\omega_0$	1

Step 5 of 8

Draw the magnitude response.

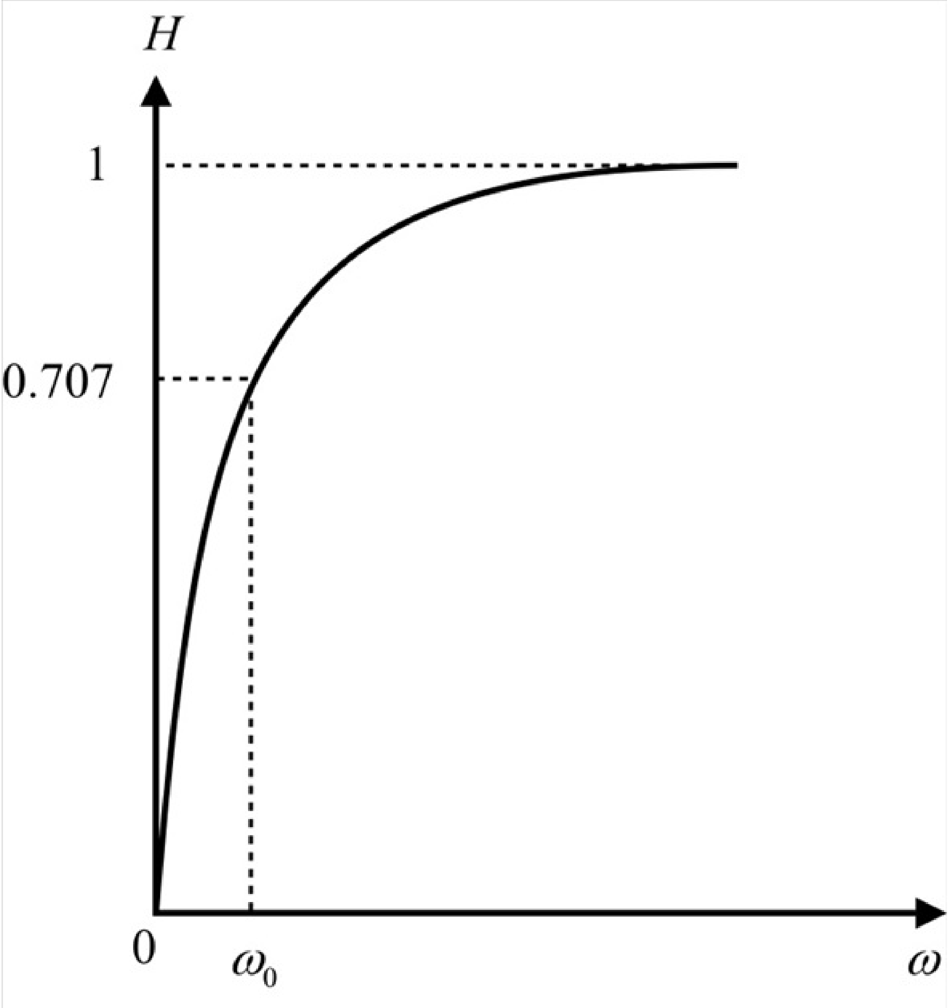


Figure 2

Step 6 of 8

Calculate the phase of the transfer function.

$$\begin{aligned}\phi &= \tan^{-1}\left(\frac{\frac{\omega}{\omega_0}}{0}\right) - \tan^{-1}\left(\frac{\frac{\omega}{\omega_0}}{1}\right) \\ &= \tan^{-1}(\infty) - \tan^{-1}\left(\frac{\omega}{\omega_0}\right) \\ &= 90^\circ - \tan^{-1}\left(\frac{\omega}{\omega_0}\right)\end{aligned}$$

Step 7 of 8

Calculate the phase at different points.

ω	ϕ
0	90°
ω_0	45°
$2\omega_0$	26.56°
$3\omega_0$	18.43°
$4\omega_0$	14.03°
$5\omega_0$	11.31°
$10\omega_0$	5.71°
$20\omega_0$	2.86°
$50\omega_0$	1.145°

Step 8 of 8

Draw the phase response.

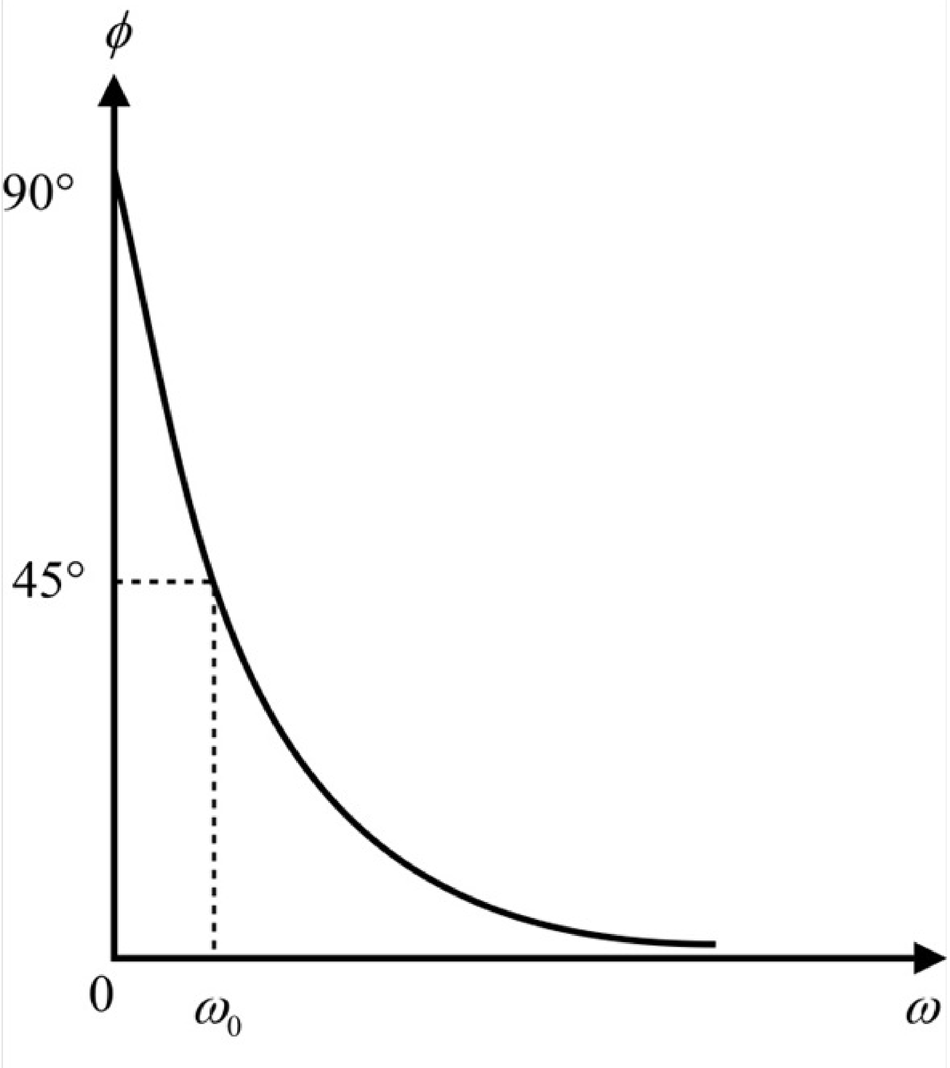


Figure 3

Thus, the frequency response is shown in Figure 2 and 3.